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Student activity

Think: Why does a business use different rates of depreciation for different types of assets?

10.7.4 Machine hour rate method

Under this method, depreciation per machine hour is calculated. The cost of the machinery after deducting the residual value, if any, is divided by the estimated working hours of the machine to find the depreciation per hour. The actual depreciation for any given period depends upon the working hours during that year. The special feature of this method is that depreciation is found directly in proportion to the actual use of the asset. Under this method life of the asset is estimated in hours and not in years. The following formula is used to determine the rate of depreciation:

$$\text{Rate of depreciation per machine hour} = \frac{\text{Original cost} - \text{Estimated scrap value}}{\text{Life of the asset in hours}}$$

Amount of depreciation = Number of machine hours used × Rate of depreciation per hour

10.7.5 Depletion method

Depletion means exhaustion of natural resources. That is, depletion means quantitative reduction in the content of assets. This is applicable to those assets that get exhausted due to extraction and exploitation. Examples: mines, oil fields, etc. Under this method, depreciation rate is calculated on the basis of the estimated quantities of the output during the whole life of the asset.

$$\text{Rate of depreciation per unit} = \frac{\text{Original cost}}{\text{Life of the asset in quantities of output}}$$

Amount of depreciation = Units of output during the year × Rate of depreciation per unit

Note: Even though it is not depreciated, it is used to write off the cost of the asset as per matching principle.

10.7.6 Annuity method

Under this method, not only the original cost of the asset but also the amount of interest on the investment is taken into account while computing depreciation. The idea of considering interest is that if the investment is made in any other asset instead of the relevant fixed asset, it would have earned a certain rate of interest. To calculate the amount of depreciation, annuity factor is used. Annuity factor can be found out from the annuity table or by using formula.

Amount of depreciation is computed as follows:

Amount of depreciation = Annuity factor × Original cost of the asset